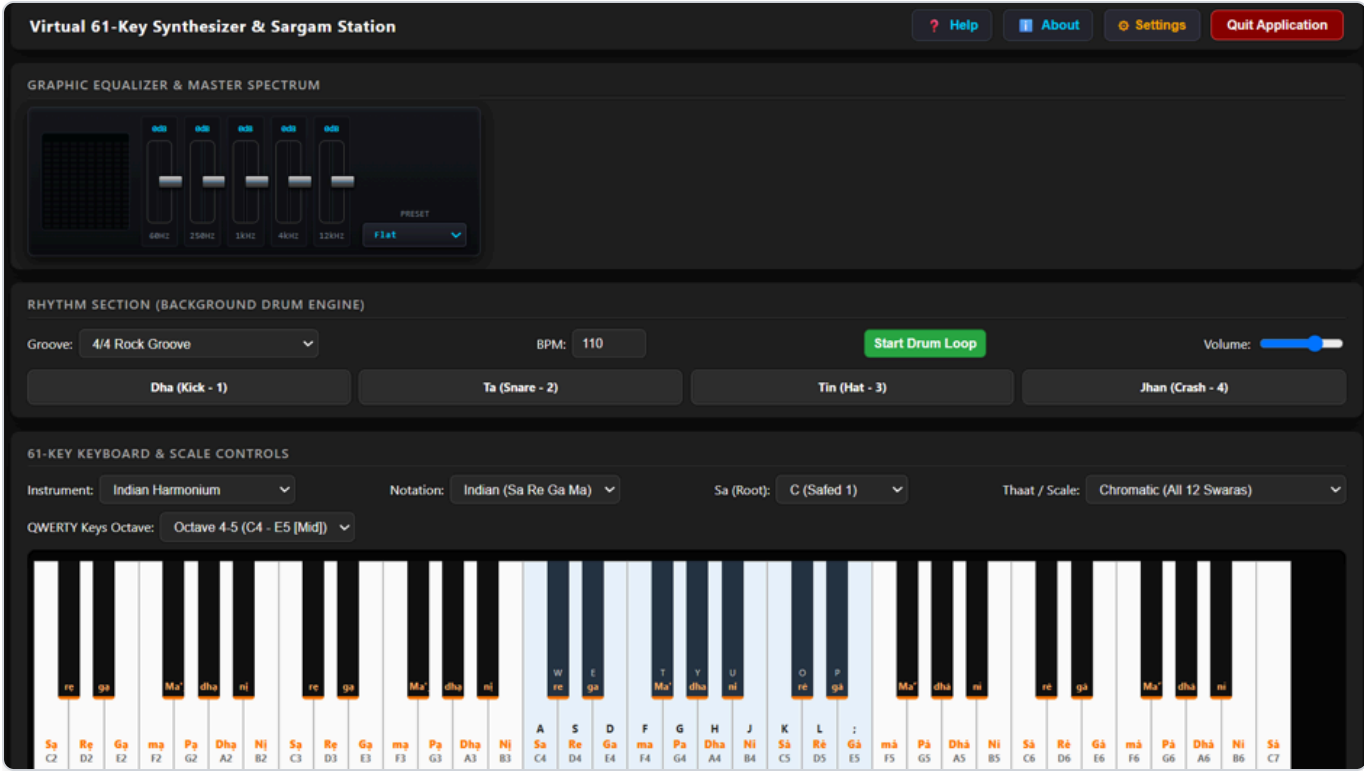


Virtual 61-Key Synthesizer & Sargam Station

User Guide & Reference Manual • Version 1.0.0 (Build 2026) • Soumyajit Chatterjee

1. Main Workstation Interface

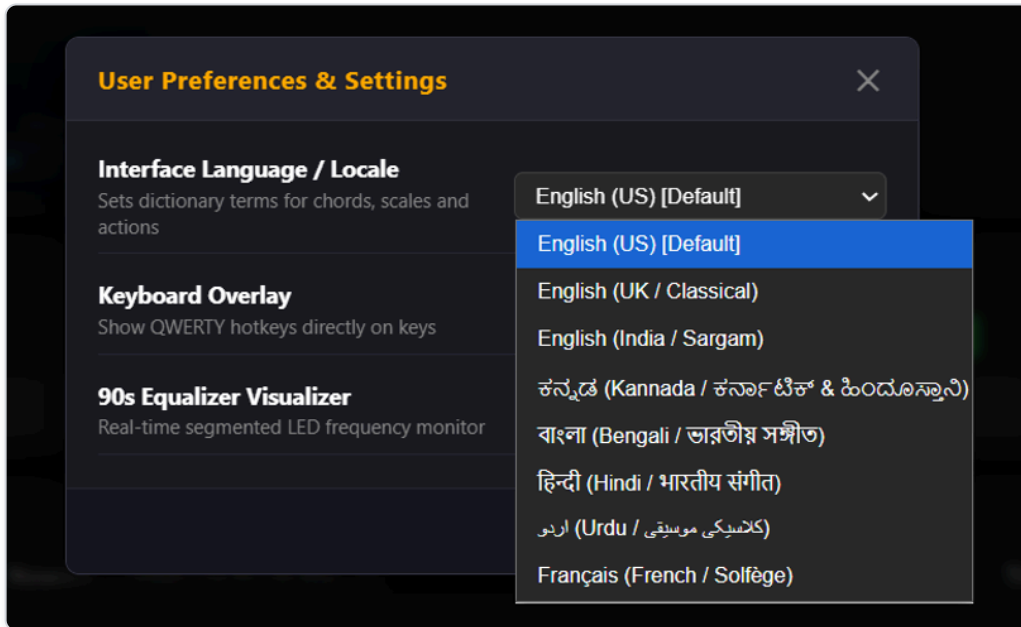
The application integrates real-time physical modeling synthesis with a multi-band graphic equalizer, backing rhythm engine, and an interactive 61-key keyboard accommodating both Indian classical Sargam and Western chromatic scales.



Section	Key Features & Functional Parameters
Graphic Equalizer	5-band sliders (60Hz, 250Hz, 1kHz, 4kHz, 12kHz), 90s-style segmented LED frequency visualizer, and sound presets (Flat, etc.).
Rhythm Section	Groove selection (4/4 Rock Groove), BPM slider/input (110 BPM), master drum loop toggle, volume fader, and 4 live percussion trigger pads.
Keyboard & Scales	Engine selector (Indian Harmonium, Piano, etc.), Notation toggle (Sa Re Ga Ma / Western), Root Sa pitch anchor, Thaata selector, and QWERTY Octave range switcher.

2. Computer Keyboard Controls & Quick Reference

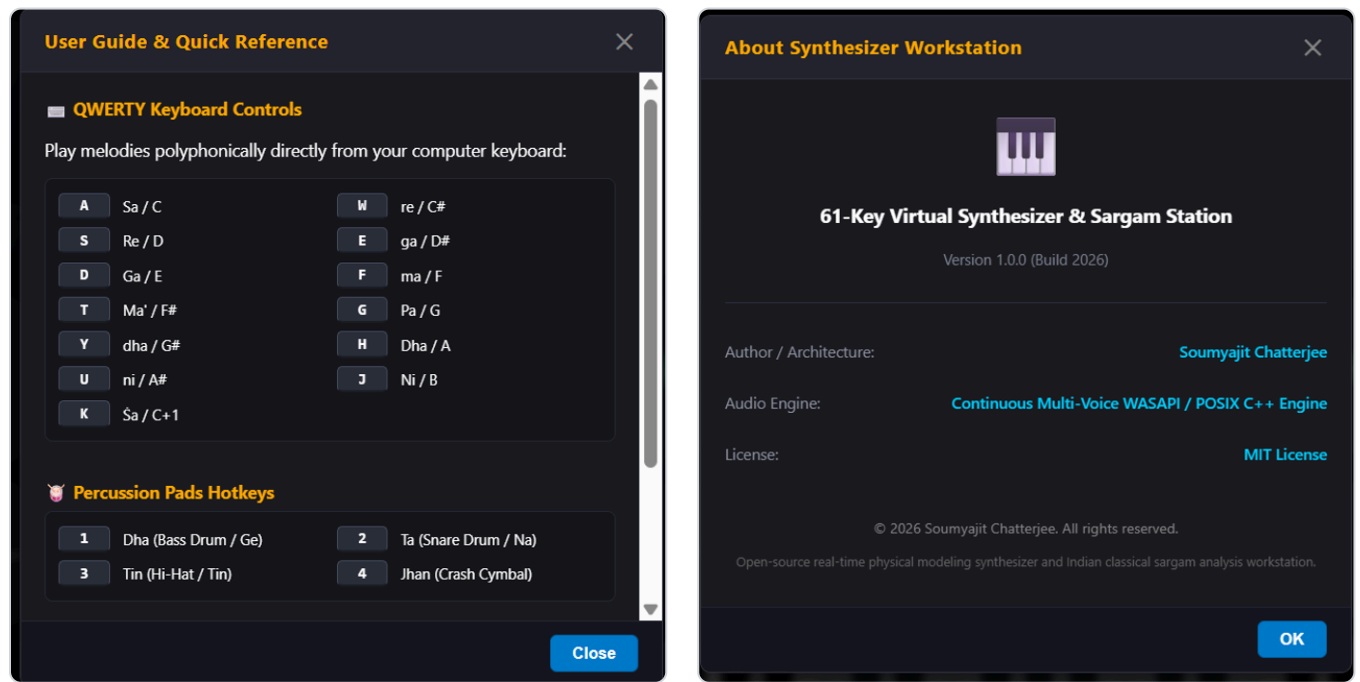
Perform live melodies polyphonically using the QWERTY keyboard. Lower-row keys operate standard natural notes (Shuddha swaras), while the upper number/letter tier triggers accidentals (Komal/Teevra swaras).



Key	Swara	Western	Key	Swara	Western	Percussion Pad	Voice
A	Sa	C	W	re	C#	1	Dha (Bass Drum / Ge)
S	Re	D	E	ga	D#	2	Ta (Snare Drum / Na)
D	Ga	E	T	Ma'	F#	3	Tin (Hi-Hat / Tin)
F	ma	F	Y	dha	G#	4	Jhan (Crash Cymbal)
G	Pa	G	U	ni	A#	Trigger live rhythmic fills directly over backing drum loops.	
H	Dha	A	K	Śa	C+1		
J	Ni	B	-	-	-		

3. User Preferences, Localization & Engine Details

The application provides extensive regional language translation for classical terminology across both Hindustani and Carnatic traditions, backed by a non-blocking audio engine.



Category	Details & Supported Options
Supported Locales	English (US) [Default], English (UK / Classical), English (India / Sargam), ಕನ್ನಡ (Kannada), বাংলা (Bengali), हिन्दी (Hindi), اردو (Urdu), and Français (French / Solfège).
Visual Toggles	Interactive QWERTY Hotkey Keycap Overlays; Segmented 90s LED Frequency Visualizer on/off.
Audio Architecture	Continuous Multi-Voice WASAPI (Windows) and POSIX C++ engine with real-time soft limiting.
Licensing	Open-source physical modeling workstation distributed under the MIT License .